

Roshan Jha

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Profile

PhD in Climate Science with **extensive experience in statistical and mathematical modeling of climate extremes and risk quantification**. Experienced in **developing probabilistic hazard models**, with a strong foundation in extreme value statistics and geospatial analysis applicable to catastrophe risk quantification. Ability to work independently and collaboratively within cross-functional teams, thrive in fast-paced commercial and research environments.

Areas of Expertise

Climate Extremes - Climate Change Scenario Analysis - Global Storm Resolving Models - Extreme Event Attribution - Geospatial Data Analysis - Statistical Modelling - Machine Learning for Earth System Modelling

Professional Experience

Postdoctoral Research Fellow ([University of St Andrews, UK](#)) 03/2025 - Present

- Investigation of **compound severe convective storm and moist heat extremes** using **Global Storm-Resolving Model (IFS and ICON)** outputs in present and near-future climate scenarios
- Developed **entrainment-based framework** using observations and climate model (CESM) to explain **compound severe convective storm and moist heat extreme** occurrence and their response to climate change

Graduate Teaching Assistant ([IIT Bombay, India](#)) 01/2022 - 05/2023

- Built **probabilistic loss-relevant frameworks** using block maxima and peaks-over-threshold analysis, GEV parameter estimation, and return level quantification directly applicable to catastrophe model hazard components
- Developed **Python-based analytical workflows for processing large climate datasets** to support hydrological modeling and flood risk assessment applications

Teaching Assistant ([IIT Bombay, India](#)) 01/2019 - 12/2019

- Facilitated open channel hydraulics experiments including flow over weirs, spillways, and hydraulic jump phenomena, building expertise in 1D/2D shallow water equation applications, Manning's roughness calibration, and flow routing techniques
- Conducted rainfall-runoff laboratory sessions using rainfall simulators and physical channel models to demonstrate overland flow generation and surface water routing

Design Engineer ([Nepal Center for Engineering and Research, Nepal](#)) 07/2017 - 07/2018

- Implemented hydraulic models using hydrodynamic simulation tool (HEC-RAS) for stormwater management in roadway infrastructure projects in United States, ensuring compliance with engineering standards and design criteria for flood risk mitigation
- Conducted **high-resolution geospatial analysis of LiDAR datasets** to generate digital elevation models (DEMs) and terrain data for flood hazard assessment and water management infrastructure design in land development projects

Education

PhD in Climate Science [Indian Institute of Technology Bombay](#) 08/2020 - 02/2025

Grade: 9.7/10 - Thesis Title: **Heatwaves in India in a changing climate: Role of atmospheric dynamics and land-atmosphere interactions**

Masters in Water Resources Engineering [Indian Institute of Technology Bombay](#) 07/2018 - 06/2020

Grade: 9.62/10 | Thesis Title: **Role of pre-monsoon irrigation on heatwaves in India**

Bachelors in Civil Engineering [TKM College of Engineering, University of Kerala](#) 08/2013 - 05/2017

Grade: 9.25/10 - Thesis Title: **Design and restoration of buildings in Munroe Island susceptible to land subsidence**

Honours & Awards

- Short Term Research Scholarship for In-Country/In Region Scholarship Holders, 2023 (Jul 2023) - [DAAD](#)
- In-Region Scholarship Programme South Asia for PhD in Climate Studies (Jul 2020) - [DAAD](#)
- DAAD Regional Scholarship for Master of Technology in Water Resources Engineering (Jun 2018) - [DAAD](#)
- 3rd Rank for 'Innovative Ideas to Post Flood Housing to rebuild Kerala' (Oct 2018) - [Government of Kerala, India](#)
- Nepal Aid Scholarship for Bachelor of Technology in Civil Engineering (Jun 2013) - [Embassy of India](#)

Skills

- **Programming:** Python (Xarray, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow), MATLAB, Bash Scripting
- **Statistical Techniques:** Extreme Value Theory (GEV, Block Maxima), Stochastic Modeling, Probabilistic Risk Assessment, Knowledge of Loss Exceedance Analysis (EP, OEP and AEP)
- **Hydraulic/Hydrological Modeling:** 1D/2D flood modeling (HEC-RAS), hydrological analysis (HEC-HMS)
- **Catastrophe Modeling Platforms:** Knowledge of loss calculation frameworks (OASIS LMF)
- **Climate Model:** Weather Research and Forecasting (WRF), Community Earth System Model (CESM)
- **Machine Learning:** Linear Regression, Decision Trees, Clustering, Artificial Neural Networks, Deep Learning Algorithms
- **Soft Skills:** Analytical Thinking, Problem-Solving, Leadership, Adaptability, Oral and Written Communication

Key Courses

Applied Statistics - Hydroinformatics - Machine Learning for Earth System Modelling - Introduction to Risk Analysis - Introduction to Ocean-Atmosphere Dynamics - Monsoons and Tropical Climate - Atmospheric Dynamics

Research Visits and Training

Short-term Research Visit (*Johannes Gutenberg-University Mainz*) **Mainz, Germany** 09/2023 - 02/2024

- Conceptualized and led the development of a research methodology to investigate pre-monsoon heatwaves in India, demonstrating strong international research experience in climate extremes and interdisciplinary environmental science
- Applied advanced **diagnostic techniques including dynamical, and process-based analyses, to elucidate the atmospheric and land-surface drivers** behind South Asian pre-monsoon heatwaves in 2022

Climate Change Adaptation Summer School (*ITT, TH Köln*) **Cologne, Germany** 12-31 Aug 2023

- Conducted **regional-scale climate risk assessment focused on flood hazards**, applying probabilistic risk frameworks and climate change scenario analysis
- Developed climate change adaptation strategy for Dodoma, Tanzania using the Climate-ADAPT European framework, GIS-based spatial analysis (QGIS), and CLIMADA catastrophe modelling platform

Workshop on Irrigation in the Earth System (*Aspen Global Change Institute*) **Aspen, USA** 04-09 Jun 2023

- Presented research on irrigation representation in land surface model (CLM), analyzing hydrological feedback mechanisms, soil moisture dynamics, and water balance under climate change scenarios
- Identified critical research gaps in irrigation-driven land surface changes affecting hydrological cycle, particularly regarding anthropogenic water cycle alterations

GeoTraining Summer School (*Goethe University*) **Frankfurt, Germany** 01-27 Sep 2019

- Executed **machine learning-based land-cover classification using Random Forest and Support Vector Machine algorithms** to derive exposure characteristics (building footprints, land-use patterns, impervious surfaces) essential for flood vulnerability modeling and catastrophe loss estimation across diverse geographies
- Developed educational framework for geospatial technologies in hazard risk assessment, incorporating DEM processing, spatial data integration, and Python workflows for hazard mapping and vulnerability analysis

Key Publications

- **Jha, R.** and Byrne, M.P. Entrainment-based framework for understanding extratropical moist heat extremes. ESS Open Archive. <https://doi.org/10.22541/essoar.177316542.20919922/v1>
- **Jha, R.**, Wirth, V. et. al. Contrasting drivers of South Asian consecutive pre-monsoon heatwaves in 2022: waveguide interaction and soil moisture depletion. JGR: Atmospheres, 130, (2025). <https://doi.org/10.1029/2024JD042376>
- **Jha, R.**, Perkins-Kirkpatrick, S.E., Singh, D. et al. Extreme terrestrial heat in 2024. Nat Rev Earth Environ 6, 234–236 (2025). <https://doi.org/10.1038/s43017-025-00661-2>
- **Jha, R.**, Mondal, A. et.al. Northward shift of pre-monsoon winds exacerbating heatwaves over India. Geophysical Research Letter, 51 (2024). <https://doi.org/10.1029/2024GL110486>
- Zachariah, M., Arulalan, T., AchutaRao, K., Saeed, F., **Jha, R.** et.al. Attribution of 2022 early-spring heatwave in India and Pakistan to climate change: lessons in assessing vulnerability and preparedness in reducing impacts. Environmental Research: Climate, 2 (2023). <https://doi.org/10.1088/2752-5295/acf4b6>
- **Jha, R.**, Mondal, A., Devanand, A. et al. Limited influence of irrigation on pre-monsoon heat stress in the Indo-Gangetic Plain. Nat Commun 13, 4275 (2022). <https://doi.org/10.1038/s41467-022-31962-5>